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Measuring Qualitative Information in Capital Markets Research: Comparison of Alternative Methodologies to Measure Disclosure Tone

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ABSTRACT: This study evaluates alternative measures of the tone of financial narrative. We present evidence that word-frequency tone measures based on domain-specific wordlists—compared to general wordlists—better predict the market reaction to earnings announcements, have greater statistical power in short-window event studies, and exhibit more economically consistent post-announcement drift. Further, inverse document frequency weighting, advocated in Loughran and McDonald (2011), provides little improvement to the alternative approach of equal weighting. We also provide evidence that word-frequency tone measures are as powerful as the Naïve Bayesian machine-learning tone measure from Li (2010) in a regression of future earnings on MD&A tone. Overall, although more complex techniques are potentially advantageous in certain contexts, equal-weighted, domain-specific, word-frequency tone measures are generally just as powerful in the context of financial disclosure and capital markets. Such measures are also more intuitive, easier to implement, and, importantly, far more amenable to replication.

Keywords: *narrative disclosure; tone; sentiment; content analysis*.

Beat *vb* 1. to strike repeatedly;. . .; 4. to overcome . . . to surpass.
—Merriam-Webster Online Dictionary

I. INTRODUCTION

Accounting standards regulate the reporting of accounting numbers, but in financial narrative, such as earnings announcements and the Management Discussion and Analysis section (MD&A) of 10-K and 10-Q filings, firms have considerable latitude over both the aspects of financial performance they choose to emphasize and the language they choose to describe performance. Two firms with identical financial accounting numbers can issue press releases that convey very different overall messages about current-period performance relative to expectations and about expected future performance. Studies in accounting and finance typically test whether the tone (i.e., sentiment) of narrative disclosure contains incremental information content, or examine factors that give rise to cross-sectional differences in the disclosure tone.

In this study, we evaluate the relative power of various approaches to measuring tone. A fundamental step in evaluating the implications of disclosure tone is to transform the narrative into a numeric value that represents the tone of the firm's message regarding performance. The most widely used approach is to count the frequency of certain words contained in the

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disclosure and compute a score based on (1) the relative frequency of a predefined list of "positive" versus "negative" words, and (2) the weights assigned to each of the positive and negative words according to a predefined weighting scheme.[1] A second general approach to measuring tone relies on machine-learning classifier algorithms that first develop classification rules based on a manually coded training dataset and then apply those rules to measure the tone in a larger dataset. Evaluation of these alternatives is particularly important given the growing use of tone measures in capital markets research, the vast amounts of qualitative financial disclosures, and the variety of approaches currently being employed to compute tone measures. In addition to the fact that some methods are likely more powerful than others, the heterogeneity in methodology limits comparability across studies.

In our evaluation of word-frequency measures, we examine four wordlists and two alternative weighting schemes used in capital markets research to compute a tone measure. The four wordlists are: (1) a wordlist developed in Henry's (2006, 2008) analysis of earnings announcements (FD wordlist); (2) a wordlist from Diction software developed and used by Roderick Hart, a specialist in politics and mass media (Diction wordlist); (3) a wordlist from the General Inquirer program developed and used by Philip Stone, a specialist in social psychology (GI wordlist); and (4) a wordlist developed in Loughran and McDonald's (2011) analysis of 10-K filings (LM wordlist).[2] We predict that the tone measures generated from the domain-specific FD and LM wordlists ($Tone_{FD}$ and $Tone_{LM}$, respectively) will outperform those generated from the general GI and Diction wordlists ($Tone_{GI}$ and $Tone_{DICTION}$, respectively) because numerous words have meanings in the context of financial disclosure that differ from the meanings assumed in GI and Diction. In general, use of a domain-specific wordlist mitigates polysemy (the capacity for a single word to have multiple meanings), one of the key generic issues in computational linguistics.

The two alternative weighting schemes used for word-frequency measures are: (1) equal weighting, and (2) an alternative weighting scheme advocated by Loughran and McDonald (2011). The method advocated by Loughran and McDonald (2011) combines a sample-dependent, inverse document frequency weighting (*idf*) with word-frequency counts to create the word-frequency-inverse document frequency measure (*wf-idf*). The *wf-idf* measure is commonly used in information retrieval algorithms (e.g., Google or other search engines) where the objective is essentially to rank a group documents according to their relevance to a specified search topic, but, arguably, the method does not logically transfer to the measurement of tone.[3] The inverse document frequency method, and others like it, down-weight words that are common in the particular sample of documents, and up-weight less common words. The most negative aspect of using *idf* weightings in tone measurement is sample dependence. With equal weighting, the word-frequency tone measure for any given document is independent of the other documents in the sample. In contrast, when *wf-idf* weighting is employed, the tone measure of each document can vary dramatically depending on which and how many other documents are in the sample. Thus, any potential improvement in the power of a document-weighted tone measure must be evaluated in the context of its diminished measurement validity and its impediment to replication.

To evaluate the alternative wordlists and weightings for word-frequency tone measures, we collect a sample of 75,599 earnings press releases issued between 2004 and 2012 from the Securities and Exchange Commission (SEC) EDGAR site. For each press release, we compute alternative tone measures. We conduct both a short-window event study and an analysis of post-earnings announcement drift.

The short-window event study tests the market reaction to unexpected earnings and to the press release's tone, where tone is measured alternately using $Tone_{FD}$, $Tone_{DICTION}$, $Tone_{GI}$, and $Tone_{LM}$. Although all four tone measures are positively associated with the market reaction to the earnings announcements, Vuong (1989) tests confirm that empirical models incorporating the $Tone_{FD}$ and $Tone_{LM}$ measures better explain the market reaction to earnings announcements than models using the $Tone_{GI}$ and $Tone_{DICTION}$ measures. Further, the economic significance of the $Tone_{FD}$ and $Tone_{LM}$ scores are greater than either the $Tone_{DICTION}$ or $Tone_{GI}$ measures. Specifically, a one standard deviation change in $\Delta Tone_{FD}$ and $\Delta Tone_{LM}$ corresponds to a 0.11 and 0.09 standard deviation change in abnormal market reaction, respectively, versus only 0.05 and 0.07 for $\Delta Tone_{DICTION}$ and $\Delta Tone_{GI}$, respectively.[4] In addition to our primary tests of the four alternative word-frequency tone measures in a short-window event study, we test the relative power of these alternative measures for smaller samples. We create randomly selected subsamples (with replacement) ranging in size from 50 to 2,000. The null is rejected 100 percent, 98 percent,

[1] Examples of research using word-frequency tone measures with equal weighting include: Henry (2006, 2008), Tetlock (2007), Tetlock, Saar-Tsechansky, and Macskassy (2008), Demers and Vega (2008), Engelberg (2008), Kothari, Li, and Short (2009), Feldman, Govindaraj, Livnat, and Segal (2010), Larcker and Zakolyukina (2012), Davis, Piger, and Sedor (2012), Davis and Tama-Sweet (2012), and Gordon, Henry, Peytcheva, and Sun (2013). Loughran and McDonald (2011) use word-frequency measures with both equal weighting and inverse document frequency weighting.

[2] The wordlist used in the General Inquirer text-processing program, the Harvard IV-d dictionary, is available at: http://www.wjh.harvard.edu/~inquirer/spreadsheet_guide.htm

[3] This method is more commonly known as "term-frequency-inverse document frequency" (*tf-idf*), but for consistency with our use of "word" versus "term" in this paper, we replace the word "term" with "word."

[4] $\Delta Tone$ equals the current-quarter earnings announcement's tone minus the prior-quarter announcement's tone, the specification of the tone measure in our main analysis.

59 percent, and 87 percent of the time in random samples of 2,000 observations using the measures $\Delta Tone_{FD}$, $\Delta Tone_{LM}$, $\Delta Tone_{DICTION}$, and $\Delta Tone_{GI}$, respectively. The relative power of the general wordlists drops considerably with smaller sample sizes. For example, the null is rejected 93 percent, 86 percent, 30 percent, and 58 percent of the time in random samples of 1,000 observations using the measures $\Delta Tone_{FD}$, $\Delta Tone_{LM}$, $\Delta Tone_{DICTION}$, and $\Delta Tone_{GI}$, respectively. These results indicate that in contexts where researchers are limited to smaller samples, Type II errors can be avoided by employing the domain-specific $\Delta Tone_{FD}$ and $\Delta Tone_{LM}$ measures.[5]

Our tests of alternative weighting methods for word-frequency tone measures compare the equal weighting method based solely on word frequencies (*wf*) and the inverse document frequency (*idf*) weighting method advocated in Loughran and McDonald (2011). We find that computing tone measures using *idf* weighting provides no improvement relative to equal weighting.

To test whether tone in press releases reflects an objective assessment of current and expected future performance or captures biased disclosures (for example, "bluffing" as an attempt to influence investors by exaggerating or offsetting the true underlying economics), we examine post-earnings announcement drift. Suppose, for example, a firm releases good news consisting of a positive earnings surprise and includes a very positive discussion of its performance. If the market initially reacts very positively, but subsequently concludes that the tone of the firm's discussion was misleadingly optimistic given the actual performance, then the firm's share price should subsequently drift downward. Alternatively, if the market initially reacts very positively and receives subsequent confirmatory evidence of good performance (i.e., subsequent in-depth analysis of actual results supports the tone of the earnings announcement), then the subsequent drift should be positive.

For our analysis of post-earnings announcement drift, we create four portfolios in which the tone measure is alternately one of the four measures studied: (1) observations with Bad News at announcement and Negative Tone, (2) observations with Bad News and Positive Tone, (3) observations with Good News and Negative Tone, and (4) observations with both Good News and Positive Tone. For the domain-specific tone measures, $\Delta Tone_{FD}$ and $\Delta Tone_{LM}$, the direction of the portfolios' drift is consistent with the direction of earnings surprise, and the level of drift is augmented by tone. Good News portfolios show positive drift, with the Good News/Positive Tone portfolios showing even higher returns than the Good News/Negative Tone portfolios. Similarly, Bad News portfolios show negative drift, with the Bad News/Negative Tone portfolios showing even lower returns than the Bad News/Positive Tone. These results indicate that the domain-specific measures are capturing attributes of qualitative information that are valued by the market.

Having evaluated alternative word-count measures of tone, we next compare word-count tone measures with the machine-learning measure used in Li (2010). Li's (2010) study uses the Naïve Bayesian machine-learning algorithm to classify each forward-looking sentence in the sample MD&A into one of four tone categories and calculates the tone measure as the average of the individual sentences' categories. Although machine-learning tone measures are a viable alternative to word-frequency measures, they have certain disadvantages for researchers. First, the initial step in classifier algorithms typically involves human coding, which—apart from being extremely labor-intensive—poses potential problems of inter-rater reliability both within a given study and across studies. Second, replication is exceedingly difficult with machine-learning algorithms not only because of variations in human coding of training data across studies, but also because an algorithm classifier, by definition, produces classifications that are specific to the particular training dataset.

To evaluate the relative efficacy of word-frequency and machine-learning tone measures, we start with the sample constructed and used in Li (2010). This sample contains measures of tone for forward-looking statements contained in the MD&A section of 10-Ks and 10-Qs filed between 1994 and 2007. We regress future earnings on the alternative tone measures and the relevant control variables defined in Li (2010). Results show that both the machine-learning tone measure and the word-frequency tone measure are positively associated with future earnings. Reported t-statistics suggest minimal differences in the power of the tests across these alternative tone measures, and Vuong's (1989) tests of differences cannot reject the null that the explanatory power of models incorporating the alternatives is equivalent. Given the marginal differences in outcomes and previously enumerated disadvantages of machine-learning tone measures, we conclude that word-frequency measures are likely preferable in most capital markets research situations examining disclosure tone. Exceptions could be in studying a type of narrative where tone has not previously been examined and, thus, for which no predefined wordlists exist, and in studies where the desired outcome is classification, in which case, the Naïve Bayesian machine-learning algorithm is particularly useful.[6]

[5] Examples of specific contexts in which capital markets researchers focus on smaller samples include subsamples of firms that exceed analysts' forecasts (identified in Brown and Caylor [2005] as perhaps the most important earnings benchmark), or a subsample of high-litigation-risk firms that have been the focus of previous capital markets research (e.g., Ajinkya, Bhojraj, and Sengupta 2005; Francis, Philbrick, and Schipper 1994).

[6] The author of the PERL module that implements the "Naïve Bayes" machine-learning algorithm, Ken Williams, writes, "It is a well-studied probabilistic algorithm often used in automatic text categorization. Compared to other algorithms (kNN, SVM, Decision Trees), it's pretty fast and reasonably competitive in the quality of its results." See: http://search.cpan.org/~kwilliams/Algorithm-NaiveBayes-0.04/lib/Algorithm/NaiveBayes.pm

This study offers several contributions to the literature. It provides a comprehensive review of four alternative wordlists that have been used for tone measures in capital markets research by examining the power of alternative tone measures to explain market reaction to earnings announcements—across varying sample sizes. This study offers both empirical and logical arguments in favor of equal weighting rather than inverse document weighting (as recommended in Loughran and McDonald [2011]) for developing word-count tone measures. This study also examines post-earnings announcement drift using the alternative tone measures to capture qualitative information. Finally, this study compares the machine-learning tone measure based on Li (2010) to the less costly and more replicable word-count tone measures, and provides evidence that both approaches produce measures that are significant in regressions of future earnings on tone. As far as we are aware, our study is the first to study the power of four alternative tone measures in a short-window event study around earnings announcements and using varying sample sizes, the first to demonstrate the relative benefits of equal weighting in word-count tone measures compared to the inverse frequency-count weighting advocated by Loughran and McDonald (2011), and the first to compare word-count measures of tone with the previously used machine-learning measures.[7]

The remainder of this paper is organized as follows. Section II describes alternative measures of the tone of disclosure in capital markets research and explains our motivation for examining these alternatives. Section III describes sample selection and defines the specific word-frequency measures, and Section IV presents descriptive statistics and results for the alternative word-frequency measures. Section V presents the comparative analysis of tone measures based on machine-learning classification versus word-frequency. Section VI concludes.

II. MEASURES OF DISCLOSURE TONE IN CAPITAL MARKETS RESEARCH

Early capital markets research assessing the market implications of qualitative financial disclosure relied on human coders making item-by-item subjective assessments of tone (Francis et al. 1994; Francis, Schipper, and Vincent 2002; Hoskin, Hughes, and Ricks 1986; Lang and Lundholm 2000). As Core (2001, 452) wrote in a review of capital markets research:

> I conjecture that researchers can substantially lower the cost of computing these [disclosure] metrics by importing techniques in natural language processing from fields like computer science, linguistics, and artificial intelligence . . . Natural language processing programs could be also used to create proxies for the "tone" of disclosure.

Today, applied computational linguistics (comprising various tasks, including content analysis and information retrieval) is a well-established literature in many disciplines. A widely used reference book on content analysis notes, "it's rare to find a text content analysis today that does *not* use computer analysis" (Neuendorf 2002; emphasis in the original).

One of the most basic tools in computational linguistics captures the content of a document with a count of the relative frequency with which particular words appear. This approach treats a document as a "bag of words," a list of the words occurring in the document and the number of times each word appears (i.e., a vector of word-frequency counts). Word-frequency counts are used to characterize the content of textual communication and, more generally, to aggregate the overall message being conveyed in the qualitative components of a disclosure.

There is a growing interest in applications of computational linguistics to disclosure in accounting and finance. A number of studies summarize disclosure tone by determining the relative frequency of "positive" versus "negative" words contained in each financial disclosure, using alternative wordlists and weighting schemes. For example, Davis et al. (2012) and Henry (2006, 2008) find a relation between earnings announcement returns and the tone of the announcement. Rogers, Buskirk, and Zechman (2011) find that sued firms' disclosures are more positive than disclosures by a matched sample of non-sued firms, and suggest that the use of more positive language increases the likelihood of being sued. Kothari et al. (2009) show that negativity in management's disclosures (proportion of negative to total words) is significantly associated with the firm's stock return volatility and analysts' forecast error dispersion, although not with a firm's cost of capital. Feldman et al. (2010) show a significant association between the tone change in the MD&A section of a firm's filings and market returns around the filing date window. Tetlock et al. (2008) find that a firm's future earnings and future stock returns are related to the tone of news stories about the firm and that the relationship is even more significant for stories that mention the word "earnings."[8] Larcker and Zakolyukina (2012), using a list of words considered "deceptive," find evidence that the level of deception in conference

[7] As well as the two lists examined in Loughran and McDonald (2011), this study examines two additional wordlists that have been used in capital market research. Beyond our inclusion of a broader set of wordlists, other important differences with their work include: our empirical results and logical arguments cautioning against the sample-dependent, *wf-idf* methodology advocated by Loughran and McDonald (2011); our use of earnings announcements to compare wordlists, as well as 10-K filings; our use of event study methodology; our demonstration of relative predictive ability of the wordlists at smaller sample sizes; our examination of post-earnings announcement drift; and our comparison of word-count-based measures with an alternative machine-learning approach.

[8] Tetlock et al. (2008) define the negative words variable as the percentage of negative words in news stories from 30 to three trading days prior to an earnings announcement, but also state that similar, but weaker, results are found using alternative definitions that incorporate positive words.

calls, measured as the frequency count of deceptive words, has predictive power with respect to future restatements. In the two subsections that follow, we examine the two major research design choices researchers must make when measuring tone: wordlists and weighting schemes.

General versus Domain-Specific Wordlists

One potential concern is that the language used in accounting disclosures tends to be somewhat specialized; essentially, a sublanguage pertaining to a specific domain of discourse. Diction and GI are two widely used wordlists that capital markets researchers adopted from other domains. Diction's wordlists were developed in the domain of political communications and are included in the commercially available software Diction 5.0. Examples of studies outside of accounting and finance that use Diction include "Media, Terrorism, and Emotionality: Emotional Differences in Media Content and Public Reactions to the September 11 Terrorist Attacks" (Cho et al. 2003) and "The Power of Leading Subtly: Alan Greenspan, Rhetorical Leadership, and Monetary Policy" (Bligh and Hess 2007). The GI wordlists have been used for more than four decades and were developed in the domain of social psychology. Examples of applications using the GI lists include "Prosody and Lexical Accuracy in Flat Affect Schizophrenia" (Alpert, Rosenberg, Pouget, and Shaw 2000) and "Some Characteristics of Genuine versus Simulated Suicide Notes" (Ogilvie, Stone, and Shneidman 1966). As is evident by the titles of these works, the GI wordlists, like Diction, have been applied in a wide variety of settings.[9]

Although it is unlikely that the use of general wordlists would lead to Type I errors, these general wordlists likely lack predictive power in a capital markets setting. The GI wordlist contains many words that are irrelevant to financial communication. For example, the GI positive wordlist includes the words "witty," "wonderful," "wondrous," and "woo," to name only a few of many words that appear irrelevant to financial communication. Additionally, numerous words are categorized as negative or positive in the general wordlists (e.g., Diction and GI) based on one meaning of the word; however, these words often have different meanings in financial disclosure. For example, GI categorizes the word "beat" as negative, but in earnings press releases, this word typically implies that companies have exceeded expectations.

An alternative to using general wordlists is to utilize domain-specific wordlists. Within the body of capital markets research, two of the tone measures we examine—$Tone_{FD}$ and $Tone_{LM}$—use wordlists that are specific to the domain of financial communication. The wordlists for $Tone_{FD}$ were developed and used in Henry's (2006, 2008) studies of earnings announcements. The wordlists for $Tone_{LM}$ were developed and used in Loughran and McDonald's (2011) study of the MD&A section of 10-Ks.

Equal versus Inverse Document Frequency Weighting

A second choice researchers must make in measuring qualitative information is the weighting of words in a wordlist. Loughran and McDonald (2011) recommend the use of inverse document frequency weighting methodologies when creating word count measures of qualitative information in financial disclosure. The authors use inverse document frequency (*idf*) weightings to produce word-frequency-inverse document frequency measures (*wf-idf*), which are widely used in the information retrieval domain, and show empirically that after applying this weighting method, the GI wordlist is at least as powerful as their own domain-specific wordlist (Loughran and McDonald 2011).

The *wf-idf* method favors words that occur in fewer documents within a sample because those words improve the ability to rank documents according to their relevance to a particular search and, thus, to obtain more precise search results. In the context of information retrieval tasks, where the objective is to sift through numerous documents and rank order those documents by their relevance to the search criteria, a document-weighting methodology such as *wf-idf* is clearly advantageous.[10] The word-frequency measure *wf* serves to capture the content of a particular document, and the *idf* weighting serves to rank order the documents by their relevance to the search term. In tasks solely involving content analysis—for example, when a document of interest has already been identified—then *idf* weighting is arguably unnecessary. In other words, studies that focus on analyzing the content of an already-identified set of documents obviously do not need to employ tools aimed at locating the documents and ranking their relevance.

[9] In the context of capital markets research, the focus is on market response to disclosure narrative. In other contexts (e.g., constituents other than investors and analysts), a more general dictionary approach could have merit.

[10] Loughran and McDonald (2011) write: "In the context of information retrieval, Jurafsky and Martin (2009, 771) note that term weighting 'has an enormous impact on the effectiveness of a retrieval system'"; however, they do not make a distinction between the task of information retrieval and the task of aggregating the qualitative information of an already-identified document. Importantly, in the referenced work, Jurafsky and Martin (2009) apply the *idf* weighting concept in the context of a retrieval system that returns an ordered set of potentially useful documents based on a users' query and explain how *idf* weighting helps to discriminate among documents.

Another feature of *idf* weighting makes its applicability for tone measurement questionable. Arguably, the commonality of a word across a sample generally would not alter the information of a given document and, thus, should not impact its weighting. With *wf-idf* weightings, where commonality determines weighting, the tone score of any given document would *always* depend on the contents of every other document in the sample. See Appendix A for an example computation of *wf-idf* and more detailed examples of problems associated with the application of *wf-idf* to tone measures.[11]

Loughran and McDonald (2011) argue that an advantage of *wf-idf* weighting is that it serves to correct for misclassified words in a wordlist. For example, the GI wordlist classifies the words "beat" and "division" as negative, but neither word should be considered negative in the context of financial disclosure. If these words appear in most of the documents in the sample, then a *wf-idf* weighting will down-weight the word significantly so that its impact on the narrative measures will become negligible. The implication is that when wordlists contain ambiguous words, *wf-idf* weighting can help to mitigate the impact. Note, however, that while *wf-idf* weighting will mitigate the impact of misclassification for very common words, it will exacerbate the impact of misclassified words that occur in a small number of documents. Additionally, *wf-idf* weighting decreases the impact of words that are correctly classified, but also appear at least once in a large number of documents across the sample.

Certain circumstances might necessitate the use of *wf-idf* weighting. As an example, the long-horizon wordlist used in Brochet, Loumioti, and Serafeim (2012) to identify management short-termism by references to the short-term versus long-term in conference call transcripts includes the word "years" among other words. Because it is likely that the word occurs at high frequencies, although not in the intended context (e.g., "as in previous years" is not a long-horizon reference), *wf-idf* weighting is likely to improve their analysis by down-weighting the word "years."

Although *wf-idf* weighting could be useful in certain circumstances, the more reliable solution when measuring tone is to exclude ambiguous and misclassified words from the wordlist to avoid the need for *wf-idf* weighting, where feasible. For example, the word "division" can be removed from a wordlist designed to capture negative financial information.

In summary, *wf-idf* weighting is an effective information retrieval methodology, but is arguably less relevant and potentially distorts tone scores. Moreover, *wf-idf* weightings are, by design, sample-dependent, which impedes replication. The advantages of equal weighting are that it is simple, transparent, and amenable to replication.[12] To evaluate the comparative efficacy of equal weighting versus *wf-idf* (hereafter referred to as "*idf*") weightings, we present results in Section IV using both equal weighting and *idf* weighting for each word-frequency tone measure.

Word-Frequency Measures versus Machine-Learning Measures

As an alternative to using wordlists and frequency counts to aggregate qualitative information, some research has used classifier algorithms that entail the following general steps: assigning a training dataset into a particular classification using some reliable mechanism (typically, manual coding), and then using the results of the manually coded data as input to a classifier algorithm that then automatically classifies a larger dataset into different categories.[13] For example, Antweiler and Frank (2004) manually classify a training dataset of 1,000 Internet bulletin board messages and use two classification algorithms (the Naïve Bayes classifier and the Support Vector Machine algorithm) to classify a sample of 1.6 million messages into one of three categories: buy, hold, or sell. Das and Chen (2007) combine classifier algorithms with a frequency-count tone measure to classify 145,000 stock message board postings for tech-sector stocks into buy, hold, or sell categories. These researchers manually classify a training set of 1,000 postings, use five different classifier algorithms to indicate how the posting should be classified, choose the classification based on the majority "vote" of the five classifiers, and refine the classification methodology by also using a measure based on the GI wordlist to eliminate ambiguous messages.

Li (2010) manually classifies 30,000 forward-looking sentences from MD&A disclosures with the help of 15 research assistants (all of whom had completed at least an intermediate accounting course). He then uses the Naïve Bayes classifier to classify a sample of 13 million forward-looking sentences as positive, negative, neutral, or uncertain. The tone measure is the average of the sentences' classifications.

Relative to word-frequency measures, classifier algorithms have several disadvantages. First, the initial step in classifier algorithms typically involves human coding, which—apart from being extremely labor-intensive—poses potential problems of inter-rater reliability both within a given study and across studies. For example, Das and Chen (2007) report that the human

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[11] See, also, Manning and Schütze (2002, Chapter 15 "Topics in Information Retrieval") for additional explanation of *idf* weightings.

[12] As Tetlock et al. (2008) write about their word-count measures, which employ equal weighting: they "are parsimonious, objective, replicable, and transparent. At this early stage in research on qualitative information, these four attributes are particularly important, and give word count measures a reasonable chance of becoming widely adopted in finance."

[13] Instead of manual coding, Balakrishnan, Qiu, and Srinivasan (2010) use one year of their data as a training set and assign the documents to one of three categories (out-performing, average, and under-performing) based on the company's actual market performance over the year following the time of their training document.

coders agreed on the classification of only 72 percent of the messages in their training sample. In contrast, word-frequency measures require far less labor and pose no problems of inter-rater reliability.

An additional disadvantage to machine-learning algorithms is that replication is far more difficult than with wordlist-based tone measurement. In addition to potential problems of inter-rater reliability of human coding of training data across studies, an algorithm classifier, by definition, produces classifications that are specific to the particular training dataset. Thus, use of a different set of documents as the training set can produce different classifications. Furthermore, different classifier algorithms also incorporate different research design choices, such as the validation process (see, e.g., Li [2010] for a discussion of validation alternatives using the Naïve Bayes method); the assumed misclassification costs and prior probability of each classification category (see, e.g., Henry [2006] for an example of assumptions in the CART classification algorithm); and the minimum frequency count for inclusion in the classifier vocabulary restrictions (see, e.g., Antweiler and Frank [2004] for an example of an assumed setting for the "prune-vocab-by-infogain" option when implementing the Naïve Bayes method in the Rainbow software package). In summary, two studies using the same classification algorithm are likely to result in different classifications of the same document. In contrast, two researchers applying the same wordlist, weighting method, and variable definition to the same document will obtain identical tone measures.

Despite disadvantages, we posit that the costs involved in a machine-learning approach might be justifiable for studying a type of narrative where tone has not previously been examined and, thus, for which no predefined wordlists exist. Further, the Naïve Bayesian machine-learning algorithm, which is a probabilistic classifier typically used in automatic text categorization, could be useful in studies where the desired outcome is classification (i.e., creation of a categorical variable). Particularly in a setting where accurate classification is of much greater importance than replicability, a machine-learning probabilistic classifier would likely be a useful tool. In our analysis, we compare the relative efficacy of word-count measures with machine-learning measures used in Li (2010).

III. RESEARCH DESIGN AND SAMPLE SELECTION: WORD-FREQUENCY TONE MEASURES

Research Design

Our objective is to evaluate tone measures constructed with alternative wordlists and weighting schemes based on their relative effectiveness in capturing the overall tone of the earnings press release narrative. As we cannot observe the "true" underlying tone—which reflects the valence of qualitative information, including management's assessment of the firm's performance—we assume semi-strong market efficiency and compare the relative information content of the alternative measures, after controlling for unexpected earnings.

Variable Construction and Research Design for Tone Measures

To conduct our tests, we first compute tone measures for each wordlist as follows:

$$Tone_{i,j} = \frac{(N^P_{i,j} - N^N_{i,j})}{(N^P_{i,j} + N^N_{i,j})} \quad (1)$$

where $Tone_{i,j}$ is the tone measure for press release *i,* based on wordlist *j* (i.e., $Tone_{i,FD}$, $Tone_{i,Diction}$, $Tone_{i,GI}$, $Tone_{i,LM}$); $N^P_{i,j}$ is the total frequency of positive words in wordlist *j,* found in document *I;* and $N^N_{i,j}$ is the total frequency of negative words in wordlist *j,* found in document *i*. For the tone measures, a purely positive press release would have a score of 1, a purely negative press release would have a score of −1, and a perfectly neutral press release would have a score of 0. The specification of the tone measure in our main analysis is change in tone ($\Delta Tone$), calculated as the current-quarter earnings announcement's tone minus the prior-quarter announcement's tone. In the regressions, all tone measures are standardized for comparability (rescaled to have a mean of 0 and a standard deviation of 1). Additional analysis explores alternative specifications for the tone measures.

We employ a short-window event study methodology to examine the relation between the market reaction to earnings announcements and the favorableness of the qualitative information in the announcements, alternately using the domain-specific and non-domain-specific wordlists. Using past research as our starting point, we begin with the following base regression (omitting fixed effects and firm observation subscripts):

$$CAR = \alpha + \beta_1 UE + \beta_2 SIZE + \beta_3 LOSS + \beta_4 \Delta Tone_j + \varepsilon \quad (2)$$

where:

CAR = cumulative abnormal returns from day *t*−1 to day *t*+1 around earnings announcement date, where abnormal returns are defined as raw returns minus value-weighted market return;

UE = (actual earnings per share − median estimated earnings per share)/beginning share price;[14]
$SIZE$ = log of market value of equity in period t;
$LOSS$ = indicator equals 1 if actual earnings per share < 0; and
$\Delta Tone_j$ = tone measure for the earnings announcement, using wordlist j, defined above.

We include a *LOSS* indicator (*LOSS* = 1 if the firm reports a loss in the current quarter) for two reasons. First, *LOSS* controls for the differential stock price reaction to earnings announcements by loss-making firms (Hayn 1995). Second, three of the wordlists, GI, DICTION, and LM, include the word "loss" as a negative word, but the FD negative wordlist does not; therefore, addition of the *LOSS* variable enables a more specific evaluation of the overlap in information between the accounting data (i.e., earnings less than zero) and the qualitative information captured by the wordlists used to measure tone. We estimate this model for each tone measure and, because the tone measures are standardized, we consider both the magnitude of the coefficient and the significance levels to assess the relative power of the alternative measures.

In addition, we create *idf*-weighted measures following Manning and Schütze (2002). Details on the *idf* measures are provided in Appendix A. For each of the alternative wordlists, the *idf*-weighted positive tone and *idf*-weighted negative tone are scaled by the total word count of the document. The overall *idf*-weighted tone measure is equal to (*idf*-weighted positive − *idf*-weighted negative)/(*idf*-weighted positive + *idf*-weighted negative). For robustness, we compute *idf* weightings using both full words and stem words.

Variable Construction and Research Design for Analysis of Post-Earnings Announcement Drift

To analyze post-earnings announcement drift, we construct four portfolios: (1) observations with Bad News at announcement (the lowest quintile of standardized unexpected earnings) and Negative Tone (lowest quintile of tone); (2) observations with Bad News and Positive Tone (bottom quintile of standardized unexpected earnings, top quintile of tone); (3) observations with Good News and Negative Tone (top quintile of unexpected earnings, bottom quintile of tone); and (4) observations both with Good News and Positive Tone (top quintile of both unexpected earnings and tone). If the tone measure is capturing the relative positivity of the qualitative information consistent with the value of that information by market participants, then we would expect the most positive drift for firms with Good News and Positive Tone and the most negative drift for firms with Bad News and Negative Tone. Where the direction of news and tone differ, we would expect less extreme drift. For example, if managers' exaggeration of good aspects of performance and/or minimizing poor aspects of performance causes investors to initially overreact to earnings news, then we would expect that extremely inconsistent tone would have an offsetting effect on post-earnings announcement drift. That is, for the subsample of firms with Bad News and Positive Tone, we would expect the drift to be flat or toward zero as investors learn that they were initially "fooled" by the tone of the press release.

Post-earnings announcement drift equals the daily mean cumulative abnormal return for each portfolio. Abnormal returns for each firm are computed as the cumulative buy-and-hold raw returns over 58 days, beginning two days after the earnings announcement, minus the firm's cumulative buy-and-hold expected returns over the same time period.[15] For each set of portfolios, we measure tone using the alternative wordlists. All variables are defined in Table 1.

Sample

Beginning in 2003, publicly listed companies are required to furnish earnings releases and similar materials to the SEC on Form 8-K (SEC 2003). We obtain earnings announcements between 2004 and 2012 from the SEC's file transfer protocol (FTP) site. We select all 8-K filings that include an earnings release identified by Item Code 2.02. There are 143,972 such filings between 2004 and 2012.[16] We then merge these data with Compustat and CRSP on the CIK (Central Index Key), which reduces the sample to 91,658 observations. The sample includes 75,599 firm-quarters after deleting observations with missing variables. Use of change in tone ($\Delta Tone$) reduces the observations to 63,357.

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[14] The median forecast is obtained from the I/B/E/S consensus file. Results are robust to use of a seasonal random walk measure of *UE*, defined as earnings per share (EPS) in quarter t minus EPS in quarter t–4, scaled by price in t–4.

[15] Each firm's risk-adjusted expected returns are derived using the Fama and French (1993) three-factor model over an estimation period of 250 trading days preceding the earnings announcement, leaving a five-day gap between the estimation period and earnings announcement day (i.e., an estimation period from day t–255 to day t–5). Results are similar for an estimation period of 150 days. In addition, for the domain-specific tone measures, the pattern of drift is similar when returns are measured as raw returns minus value-weighted market returns and when portfolios are formed on the basis of announcement date CAR (cumulative abnormal returns) rather than unexpected earnings.

[16] This is not an all-inclusive list of earnings press releases. One shortcoming of the SEC filings is that the Item codes are unaudited and a large number of firms miscode their 8-K filings. We only include firms that properly classified their 8-K filing as an earnings release. Note, also, that the new 8-K coding did not take effect until August 2004, so our sample effectively starts in August 2004.

TABLE 1

List of Variables

Variable	Definition
CAR	Cumulative abnormal returns from day $t-1$ to day $t+1$ around earnings announcement date, winsorized at 1st and 99th percentiles.
UE	Unexpected earnings per share (Actual earnings per share − Median estimated earnings per share)/beginning share price, winsorized at 1st and 99th percentiles.
SIZE	Log of market value of equity, calculated as number of common shares times share price, in millions of dollars, and winsorized at 1st and 99th percentiles.
NUM_WORDS	Count of total words in the disclosure, winsorized at 1st and 99th percentiles.
$Tone_{FD}$	(*POSITIVE* − *NEGATIVE*)/(*POSITIVE* + *NEGATIVE*) *where POSITIVE and NEGATIVE* refer to the word count frequency based on the positive and negative words in the FD wordlist, respectively.
$Tone_{DICTION}$	(*POSITIVE* − *NEGATIVE*)/(*POSITIVE* + *NEGATIVE*) *where POSITIVE and NEGATIVE* refer to the word count frequency based on the positive and negative words in the Diction wordlist, respectively.
$Tone_{GI}$	(*POSITIVE* − *NEGATIVE*)/(*POSITIVE* + *NEGATIVE*) *where POSITIVE and NEGATIVE* refer to the word count frequency based on the positive and negative words on the General Inquirer wordlist, respectively.
$Tone_{LM}$	(*POSITIVE* − *NEGATIVE*)/(*POSITIVE* + *NEGATIVE*) *where POSITIVE and NEGATIVE* refer to the word count frequency based on the positive and negative words on the Loughran-McDonald wordlist, respectively.
$\Delta Tone_j$	Change in tone. For each wordlist (FD, Diction, GI, and LM), current quarter earnings announcement's tone minus prior quarter announcement's tone.
LOSS	Indicator equals 1 if actual earnings per share < 0.

We process the 8-K filings using PERL scripts. We first strip out the earnings release from any other items included with the 8-K filings. We also remove any HTML or SGML tags. Next, the PERL script conducts a frequency count for words in each of the four sets of wordlists (FD, DICTION, GI, and LM). We "preprocessed" the GI wordlist to account for word "stems." For example, the entry "decrease" on the GI list is a stem (the primary lexical root of the word), which is the singular form of the noun and the infinitive form of the verb. A frequency count of the word, in all its forms, can be obtained either by instructing the text-processing program to include all words that contain the stem or, alternatively, by expanding the list itself to include all forms of the word—"decrease, decreases, decreased, decreasing." We use the second approach because the other wordlists include complete words rather than stems. One reason for not using stemming in a wordlist is that not all forms of a word are necessarily equally indicative of positive or negative information, particularly in a specific domain of discourse. For example, in earnings announcements, the word "record" is positive because it is most frequently used as an adjective, as in "record sales." In contrast, the word "recorded" would typically be used as the past tense of the verb without positive or negative implications. The FD wordlist includes a total of 93 negative words and 188 positive words. The wordlists from Diction used in accounting research to measure positive and negative include 914 negative words and 697 positive words. The wordlists from GI contain 3,699 negative words and 2,557 positive words. The negative LM wordlist contains 2,337 words, and the positive LM wordlist contains 353 words.

IV. DESCRIPTIVE STATISTICS AND RESULTS: WORD-FREQUENCY TONE MEASURES

Descriptive statistics for the tone scores of our sample of 63,357 earnings announcements are shown in Table 2. (Statistics for the tone measures are shown prior to standardization.) The mean and median values of $Tone_{FD}$ (mean = 0.400 and median = 0.440), $Tone_{GI}$ (mean = 0.334 and median = 0.337), and $Tone_{DICTION}$ (mean = 0.285 and median = 0.312) are significantly higher (i.e., more positive) than the mean and median values of and $Tone_{LM}$ (mean = −0.024 and median = −0.043).[17] Prior research on qualitative financial disclosure indicates that voluntary disclosure tends to be positive (Abrahamson and Amir 1996; Rutherford 2005.) Thus, we would expect that the average values of the tone measures for earnings press release to be

[17] The likely reason that the mean and median of the tone score using the LM wordlists are negative is that the LM wordlists contain nearly seven times more negative words than positive. In contrast, for the other three wordlists, the number of negative words is much closer to the number of positive words. Specifically, the ratio of Max(negative words, positive words) to Min(negative words, positive words) is 2 or below for the FD, Diction, and GI wordlists.

TABLE 2
Sample (n = 63,357)

Panel A: Descriptive Statistics

Variable	Mean	Median	Std. Dev.	Minimum	Maximum
CAR (%)	0.148%	−0.020%	8.136%	−24.563%	25.770%
UE (%)	0.003%	0.050%	1.299%	−8.037%	5.000%
LOSS	0.184	NA	NA	0	1
SIZE	6.844	6.751	1.678	3.216	11.196
NUM_WORDS	2,518	2,242	1,279	694	7,580
$Tone_{FD}$	0.400	0.440	0.298	−1	1
$Tone_{DICTION}$	0.285	0.312	0.362	−1	1
$Tone_{GI}$	0.334	0.337	0.156	−0.556	0.917
$Tone_{LM}$	−0.024	−0.043	0.345	−1	1
$\Delta Tone_{FD}$	−0.003	0.000	0.203	−1.500	2.000
$\Delta Tone_{DICTION}$	0.007	0.000	0.227	−2.000	2.000
$\Delta Tone_{GI}$	−0.001	−0.001	0.092	−0.711	0.690
$\Delta Tone_{LM}$	−0.002	0.000	0.232	−1.750	2.000

Panel B: Frequency of Observations by Year

File Date Year	Frequency	Percent
2004	137	0.22
2005	7,497	11.83
2006	7,953	12.55
2007	6,324	9.98
2008	6,285	9.92
2009	8,330	13.15
2010	8,925	14.09
2011	8,973	14.16
2012	8,933	14.10
Total	63,357	100.00

For regressions, all tone variables are standardized (rescaled to have a mean of 0 and a standard deviation of 1). This table reports non-standardized statistics. Sample selection period begins in August 2004, when the coding of 8-K filings as an earning press release took effect. Beginning in 2003, the SEC requires that earnings press releases must, if issued, be filed with the SEC on Form 8-K (SEC 2003).
Variables are defined in Table 1.

greater than zero. The consistency between the expected bias toward positivity and the higher levels of $Tone_{FD}$ and $Tone_{GI}$ score suggests that these two measures have more face validity than $Tone_{DICTION}$ or $Tone_{LM}$ in capturing the tone of voluntary financial disclosure.

Three of the tone measures—$Tone_{FD}$, $Tone_{DICTION}$, and $Tone_{LM}$—have a maximum value of 1.00, indicating a purely positive tone and a minimum value of −1.00, indicating a purely negative tone, but $Tone_{GI}$ has a maximum (minimum) of 0.917 (−0.556). Table 2 also presents descriptive statistics for the change in tone, with means and medians close to zero for all measures. Table 2, Panel B reports sample frequencies by years. With the exception of 2004, yearly sample sizes range from 6,285 in 2008 to 8,973 in 2011. The small sample size in 2004 (137) is primarily the result of the timing of the SEC rule change requiring firms to file earnings announcements in an 8-K. In addition, our change specification requires prior-quarter earnings announcements.

Table 3 reports the correlation between each of the tone measures and the following: cumulative abnormal returns around the earnings announcement date (*CAR*), unexpected earnings (*UE*), the size of the announcing firm based on market value of equity (*SIZE*), and the length of the earnings announcement measured as the word count (*NUM_WORDS*). Comparing the tone measures, we find that *CAR* is more highly correlated with $\Delta Tone_{FD}$ than with the other tone measures.

TABLE 3

Correlation Matrix
Pearson Correlation Coefficients
(n = 63,357)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
1. *CAR*												
2. *UE*	**0.221**											
3. *LOSS*	**−0.115**	**−0.310**										
4. *SIZE*	0.009	**0.081**	**−0.342**									
5. *NUM_WORDS*	−0.006	**−0.012**	**−0.079**	**0.322**								
6. $Tone_{FD}$	**0.090**	**0.100**	**−0.212**	**0.141**	−0.008							
7. $Tone_{DICTION}$	**0.061**	**0.098**	**−0.371**	**0.229**	−0.001	**0.377**						
8. $Tone_{GI}$	**0.048**	**0.063**	**−0.112**	**−0.039**	**−0.122**	**0.295**	**0.311**					
9. $Tone_{LM}$	**0.083**	**0.126**	**−0.237**	**0.096**	**−0.160**	**0.464**	**0.522**	**0.343**				
10. $\Delta Tone_{FD}$	**0.118**	**0.073**	**−0.017**	0.000	**0.026**	**0.344**	**0.074**	**0.099**	**0.118**			
11. $\Delta Tone_{DICTION}$	**0.068**	**0.068**	**−0.092**	−0.010	−0.006	**0.073**	**0.306**	**0.102**	**0.154**	**0.222**		
12. $\Delta Tone_{GI}$	**0.087**	**0.079**	**−0.060**	0.002	**−0.027**	**0.121**	**0.118**	**0.292**	**0.144**	**0.338**	**0.366**	
13. $\Delta Tone_{LM}$	**0.111**	**0.097**	**−0.061**	−0.008	**−0.015**	**0.118**	**0.149**	**0.120**	**0.336**	**0.350**	**0.478**	**0.425**

Coefficients in bold are significant at $p < 0.01$.
Variables are defined in Table 1.

Empirical Tests of the Relation between Market Reaction and Alternative Measures of the Qualitative Information in Earnings Announcements

Table 4 presents the results of estimating Equation (2).[18] For the estimation, each of the tone variables is standardized. The coefficients estimated for each of the tone measures are positive and significant. The magnitude of the coefficient on $\Delta Tone_{FD}$ is greater than the coefficient on the other three scores. Results of Vuong (1989) tests, shown in Table 4, Panel B, indicate that the models incorporating the domain-specific $\Delta Tone_{FD}$ or $\Delta Tone_{LM}$ scores have greater predictive ability than those using either $\Delta Tone_{DICTION}$ or $\Delta Tone_{GI}$ scores, $\Delta Tone_{FD}$ outperforms $\Delta Tone_{LM}$, and $\Delta Tone_{GI}$ outperforms $\Delta Tone_{DICTION}$.[19]

When the regressions are estimated using the level of tone rather than change in tone, the comparative results are similar.[20] We repeat the regressions using change in analysts' earnings forecasts for the quarter following the earnings announcement (Q+1) as the dependent variable. Untabulated results show that change in analysts' forecast is positively and significantly associated with each of the tone measures. In tests of comparative explanatory power, results are qualitatively similar to the main results in that models incorporating $\Delta Tone_{FD}$ have greater predictive ability than models incorporating any of the other three measures. The relative predictive ability of $\Delta Tone_{DICTION}$ and $\Delta Tone_{GI}$ is equivalent.

We examine whether the market response to tone has changed in more recent years in light of the increase in algorithmic trading, which often incorporates computational linguistics.[21] Untabulated results from estimating the regressions in Table 4 on a year-by-year basis show that comparative power of the different measures has not changed. Specifically, the significance (t-statistics) of the coefficients on tone variables using FD and LM wordlist measures (with FD dominating in seven of the eight years) is greater than Diction and GI. While the most recent three years (2010–2012) show a year-on-year increase in the magnitude and significance of the coefficients for some of the tone measures, the trend is not consistent across the sample period.

[18] Reported results include quarter fixed effects in regressions, with robust standard errors clustered by firm and quarter. Untabulated results using year and firm fixed effects yield the same conclusions.

[19] When tone measures are constructed as $(POS - NEG)/NUM\_WORDS$, untabulated results show that $\Delta Tone_{FD}$ and $\Delta Tone_{LM}$ scores still have greater predictive ability than those using either $\Delta Tone_{DICTION}$ or $\Delta Tone_{GI}$ scores. $\Delta Tone_{GI}$ and $\Delta Tone_{DICTION}$ are equivalent, and within the domain-specific measures, $\Delta Tone_{FD}$ and $\Delta Tone_{LM}$ are equivalent.

[20] We also undertake an analysis including multiple tone measures in the model. Untabulated results confirm that no single measure subsumes all the others.

[21] We thank an anonymous reviewer for this suggestion.

TABLE 4

Regression of Cumulative Abnormal Returns on Unexpected Earnings and Tone Measures
(n = 63,357)

$$CAR = \alpha + \beta_1 UE + \beta_2 SIZE + \beta_3 LOSS + \beta_4 \Delta Tone_j + \varepsilon$$

Panel A: Regression of Abnormal Returns on Unexpected Earnings and Tone Measures, Controlling for Loss-Making Companies

	Coefficient (t-stat)	Coefficient (t-stat)	Coefficient (t-stat)	Coefficient (t-stat)
Intercept	0.015***	0.014***	0.015***	0.015***
	(6.73)	(6.59)	(6.79)	(6.83)
UE	1.228***	1.262***	1.251***	1.232***
	(16.18)	(16.17)	(16.19)	(16.24)
SIZE	−0.001***	−0.001**	−0.001**	−0.001**
	(−3.54)	(−3.34)	(−3.35)	(−3.21)
LOSS	−0.014***	−0.012***	−0.013***	−0.013***
	(−9.57)	(−8.58)	(−9.11)	(−8.89)
$\Delta Tone_{FD}$	0.009***			
	(15.67)			
$\Delta Tone_{DICTION}$		0.004***		
		(9.36)		
$\Delta Tone_{GI}$			0.006***	
			(11.92)	
$\Delta Tone_{LM}$				0.007***
				(11.67)
Quarter Fixed Effects	Yes	Yes	Yes	Yes
Adjusted R^2	6.55%	5.67%	5.90%	6.18%

Panel B: Results of Vuong's Test

Competing Models' Tone Measure	Vuong's Z-statistic	p-value
$\Delta Tone_{FD}$ versus $\Delta Tone_{DICTION}$	10.862	<0.001
$\Delta Tone_{FD}$ versus $\Delta Tone_{GI}$	8.030	<0.001
$\Delta Tone_{FD}$ versus $\Delta Tone_{LM}$	4.392	<0.001
$\Delta Tone_{DICTION}$ versus $\Delta Tone_{GI}$	−4.440	<0.001
$\Delta Tone_{DICTION}$ versus $\Delta Tone_{LM}$	−8.672	<0.001
$\Delta Tone_{GI}$ versus $\Delta Tone_{LM}$	−4.285	<0.001

***, **, * Indicate significance at < 0.001, < 0.01, and < 0.05, respectively, two-tailed.
Tone variables are standardized (rescaled to have a mean of 0 and a standard deviation of 1). Regressions use robust standard errors, clustered by GVKEY and quarter. Results of Vuong's test are reported in Panel B.
Variables are defined in Table 1.

Tone Measures with *idf* Weightings

The results described above are based on tone measures using equal weighting. In Table 5, we report regressions for each tone measure computed under both equal weighting and *idf* weighting. As shown, *idf* weighting shows no improvement over equal weighting. Our findings differ from Loughran and McDonald (2011), who find a dramatic improvement in the effectiveness of the GI wordlist when they apply *idf* weighting to 10-K disclosures; however, their measures are based on only on negative words. In untabulated results, similar to Loughran and McDonald (2011), we do find that *idf* weighting for negative measures (ΔNEG) yields statistically significant improvements over equal weighting for ΔNEG_{GI}, although the economic significance is quite modest.

TABLE 5

Comparison of Tone Measures with Equal-Weighing to Tone Measures with *idf* Weighting (n = 63,357)

$$CAR = \alpha + \beta_1 UE + \beta_2 SIZE + \beta_3 LOSS + \beta_4 \Delta Tone_j + \varepsilon$$

	FD		DICTION		GI		LM	
	Equal Coefficient (t-stat)	*idf* Coefficient (t-stat)	Equal Coefficient (t-stat)	*idf* Coefficient (t-stat)	Equal Coefficient (t-stat)	*idf* Coefficient (t-stat)	Equal Coefficient (t-stat)	*idf* Coefficient (t-stat)
Intercept	0.015***	0.014***	0.014***	0.015***	0.015***	0.016***	0.015***	0.016***
	(6.73)	(6.43)	(6.59)	(6.77)	(6.79)	(7.20)	(6.83)	(7.04)
UE	1.228***	1.245***	1.262***	1.269***	1.251***	1.266***	1.232***	1.253***
	(16.18)	(16.08)	(16.17)	(16.05)	(16.19)	(16.12)	(16.24)	(16.10)
SIZE	−0.001***	−0.001**	−0.001**	−0.001**	−0.001**	−0.001**	−0.001**	−0.001**
	(−3.54)	(−3.45)	(−3.34)	(−3.46)	(−3.35)	(−3.47)	(−3.21)	(−3.39)
LOSS	−0.014***	−0.013***	−0.012***	−0.013***	−0.013***	−0.013***	−0.013***	−0.013***
	(−9.57)	(−9.40)	(−8.58)	(−9.42)	(−9.11)	(−9.54)	(−8.89)	(−9.56)
$\Delta Tone_j$	0.009***	0.007***	0.004***	0.004***	0.006***	0.004***	0.007***	0.006***
	(15.67)	(14.00)	(9.36)	(8.63)	(11.92)	(10.17)	(11.67)	(11.47)
Quarter Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	6.55%	6.26%	5.67%	5.64%	5.90%	5.71%	6.18%	5.96%

***, **, * Indicate significance at < 0.001, < 0.01, and < 0.05, respectively, two-tailed.
Robust standard errors are clustered by GVKEY and quarter. $\Delta Tone_j$ refers to the four alternative wordlists (FD, DICTION, GI, and LM). Tone variables are standardized (rescaled to have a mean of 0 and a standard deviation of 1).
All other variables are defined in Table 1.

Alternative Measures of Unexpected Tone

Our main findings suggest that unexpected tone using a random walk model (prior-quarter tone) to measure expected tone is a significant improvement over using tone levels. For each tone measure, untabulated results show that using change in tone provides greater explanatory power in our main regressions than using level of tone. The single-quarter differencing to derive the change in tone appears to reduce noise in the tone measures.

We also consider alternative approaches to derive unexpected tone. As one alternative, we calculate a measure of unexpected tone as the difference between actual tone and a moving average of the prior four quarters, which can potentially provide a more stable "base case." Untabulated findings indicate that the moving-average measure does not increase explanatory power beyond the single-quarter change in tone.

Robustness of Results to Inclusion of Additional Financial Variables

We view the tone measures as a summary of the firm's qualitative performance assessment. However, if these word-based measures do not provide incremental information beyond key financial metrics, then such measures are redundant and unnecessary. Prior research suggests that qualitative disclosure does provide incremental information beyond financial metrics (e.g., Henry 2006). We examine the eight key financial variables used in Piotroski (2000).[22] Results (untabulated) are robust to including these additional financial variables. Specifically, all tone measures are positive and significant, and the relative explanatory power of the different measures is substantively the same as described above.

[22] All variables are defined as in Piotroski (2000). ROA equals net income before extraordinary items, scaled by total assets at the beginning of the year. Gross margin equals net sales less cost of goods sold, scaled by net sales. Liquidity is measured as the current ratio, i.e., current assets divided by current liabilities. Leverage is measured as debt-to-assets, i.e., total debt divided by average total assets. Asset turnover equals net sales divided by average total assets. Accruals are measured as net income before extraordinary items less cash flow from operations, scaled by beginning assets. Note, however, that some of the financial items may not be available at the time of the earnings announcements, but only later when the 10-Q or 10-K is filed.

Empirical Tests of the Relative Power of Alternative Tone Measures for Smaller Samples

In the previous sections, we show that in a regression of abnormal returns on unexpected earnings and alternative tone measures, the coefficients on $\Delta Tone_{FD}$ and $\Delta Tone_{LM}$ are larger in magnitude than the corresponding coefficients on $\Delta Tone_{DICTION}$ and $\Delta Tone_{GI}$. In addition, the explanatory power of regressions including $\Delta Tone_{FD}$ and $\Delta Tone_{LM}$ exceeds that of the regressions using either of the other scores. We infer from this that the $\Delta Tone_{FD}$ and $\Delta Tone_{LM}$ scores are statistically more powerful than $\Delta Tone_{DICTION}$ and $\Delta Tone_{GI}$ when aggregating qualitative information in the press release. In this section, we extend our analysis by evaluating the relative statistical power of the four measures to reject the null (of no incremental information content in measures of qualitative information) when sample sizes are relatively small.

As in the previous sections, we employ a short-window event study methodology to examine the relation between the market reaction to earnings announcements and alternative tone measures, using both equal weighting and *idf* weighting for both tone change and level. We explore the relative power of the four measures using randomly selected subsamples of varying sizes ranging from 50 to 2,000. We obtain parameter estimates from 250 regressions on randomly selected subsamples (with replacement) for each sample size. Figure 1, Panel A plots the rejection rates at $p < 5$ percent for the coefficients on each of the $\Delta Tone$ measures constructed with equal weighting. Panel A shows that the rejection rates converge to almost 100 percent for $Tone_{FD}$ and $Tone_{LM}$ as the sample size increases to 2,000, and the rejection rates for the two other tone measures are lower.

Figure 1, Panel B presents results using *idf*-weighted tone measures. For the *idf*-weighted measures, a separate *idf* weighting is calculated for each of the 1,500 different subsamples, simulating the process that a researcher would take. Recall that the weight assigned to any word depends on whether and how frequently it appears in each document in a given sample and, therefore, *idf* weights are computed for each random sample. A comparison of Figure 1, Panel A to Panel B suggests that *idf* weighting does little to improve the power of the alternative tone measures. For example, at the 500-observation level, the rejection rate for the Diction tone measure increases from around 16 percent in Panel A to 20 percent in Panel B, but at the 2,000-observation level, the power of the *idf*-weighted measure is the same or worse than the equal-weighted measure, for all four tone alternatives.

Figure 1, Panel C presents the rejection rates for tone measures using the level rather than change specification. While the rejection rates rise to 95 percent for the FD wordlist and 73 percent for the LM wordlist at the 2,000-observation level, the rejection rates for the other two word lists remain quite low. This suggests that the power of general wordlists to capture qualitative information—using levels specification for measuring tone—is relatively low in this financial disclosure setting and, if used, would substantially increase the likelihood of Type II errors in small sample sizes or where the effects of tone measures are smaller. For example, even at sample sizes of 2,000, rejection rates for both general wordlists remain at or below 36 percent.

Figure 1, Panel D presents the rejection rates for tone measures using the level rather than change specification, constructed using *idf* weighting. A comparison of Figure 1, Panel C to Panel D again suggests that *idf* weighting does little to improve the power of the alternative tone measures, and the effect varies. For example, rejection rates for the *idf*-weighted GI tone measure are equal to or greater than rejection rates for the equal-weighted GI tone measure for the 50, 100, 250, and 500 sample sizes, but lower than rejection rates for the equal-weighted GI tone measure for the 1,000 and 2,000 sample sizes.

Overall, the evidence in Figure 1 is consistent with domain-specific wordlists being more powerful in the context of financial disclosure. Moreover, the evidence suggests that using the change specification rather than the level specification of tone is important, particularly for the non-domain-specific wordlists.

Analysis of the Key Reasons for the Differences in Tone Measures

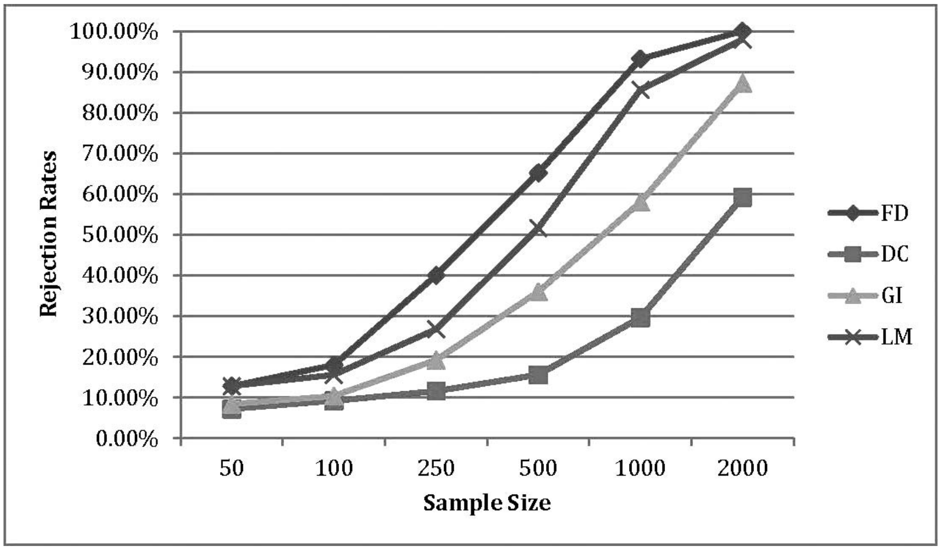
To understand the reasons for the differences in the tone scores, we identify and analyze earnings announcements for which the different wordlists yield the greatest measurement differences (specifically, where $Tone_{FD}$ is very different from $Tone_{GI}$). For each press release, we compute a score divergence as $Tone_{FD}$ minus $Tone_{GI}$. If the score divergence is a negative (positive) number, then the FD wordlist measures the qualitative information of the press release as more negative (positive) than the GI wordlist. We identify the top 5 percent of press releases, ranked by divergent scores—one group of press releases measured as more negative using the FD wordlist and one group of press releases measured as more positive using the FD wordlist. The most divergent releases are then examined by group.

For the group of press releases measured as much more negative using the FD wordlist, Panel A of Table 6 lists the most frequently occurring words appearing in the GI positive wordlist, but not in the FD positive wordlist, and Panel B of Table 6 lists the most frequently occurring words appearing in the FD negative wordlist, but not in the GI negative wordlist.[23] Across

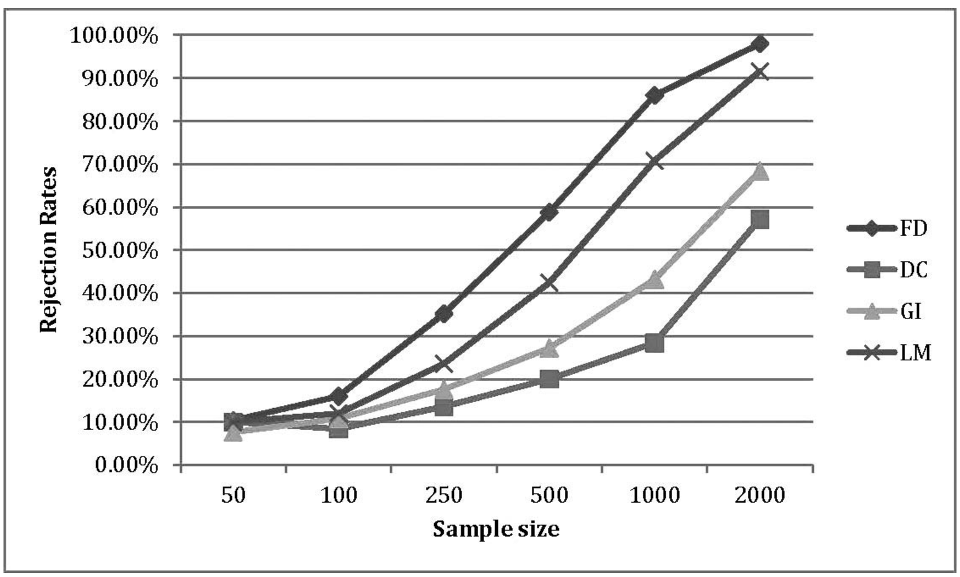
[23] We limit the list to the most frequently occurring, non-overlapping words because, as is generally the case in computational linguistics, the distribution of occurrences of words is highly skewed (i.e., most words are rare). In addition, we aggregate by stem word for presentation purposes. Because our unit of analysis is the wordlists used in prior research, we do not examine whether exclusion or inclusion of a particular word would produce a misleading tone measure.

FIGURE 1
Comparison of the Power of the Tone Measures for 250 Randomly Selected Samples

Panel A: Rejection Rates (β_4) for Δ*Tone* Measures Constructed with Equal Weighting



Panel B: Rejection Rates (β_4) for Δ*Tone* Measures Constructed with *idf* Weighting



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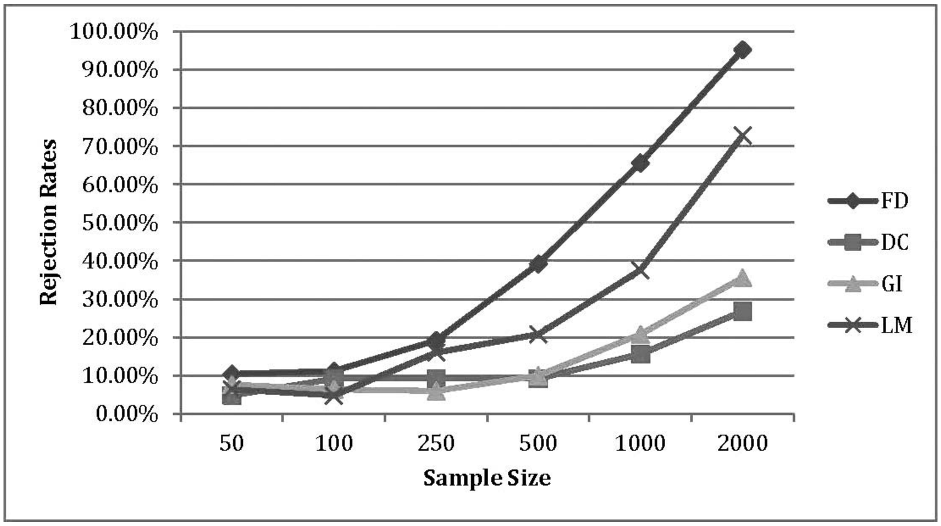
both panels, the word that occurs in the largest number of this group of press releases is the word "share," which the GI list includes as a positive word. Used in certain ways, the word "share" has positive connotations; however, in the domain of financial disclosure, it can be expected that the word "share" refers to a certificate of ownership of a company and, thus, has neither positive nor negative connotations.

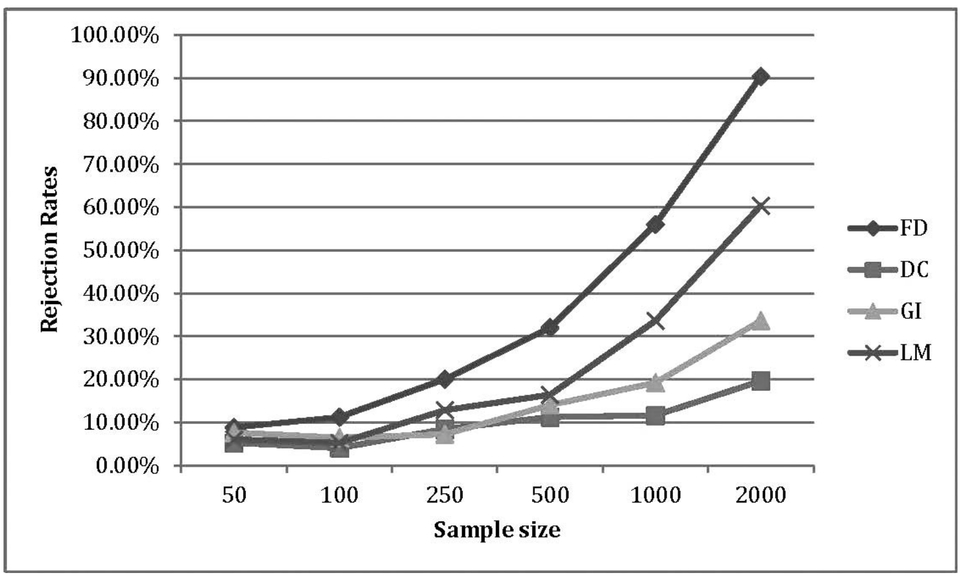
For the group of press releases measured as much more positive using the FD wordlist, Panel A of Table 7 lists the most frequently occurring words appearing in the FD positive wordlist, but not in the GI positive wordlist, and Panel B of Table 7 lists the most frequently occurring words appearing in the GI negative wordlist, but not in the FD negative wordlist. Across

FIGURE 1 (continued)

Panel C: Rejection Rates (β_4) for *Tone* Measures (Level) Constructed with Equal Weighting



Panel D: Rejection Rates (β_4) for *Tone* Measures (Level) Constructed with *idf* Weighting



For randomly selected samples of 50, 100, 250, 500, 1,000, and 2,000 observations, the following equation was estimated: $CAR = \alpha + \beta_1 UE + \beta_2 SIZE + \beta_3 LOSS + \beta_4 \Delta Tone_j + \varepsilon$ where all variables are defined in Table 1, and $Tone_j$ refers to the four alternative tone scores.
The full-color version of Figure 1 is available as a downloadable PDF file, see Appendix B.

both panels, the most frequently occurring stem word is "increas," which the FD list includes as a positive word, but the GI wordlist does not.[24] In untabulated counts based on complete words rather than stems, the most frequently occurring words are "increase" and "increased."

[24] Henry (2008, 401) undertakes an affect disambiguation analysis, which supports inclusion of directional words in the FD wordlist, such as the word "increase" in the positive wordlist. She documents, for example, that 83 percent of the non-neutral words occurring immediately to the left of the word "increased" (i.e., its L1 to L3 collocates) in earnings announcements referred to a generally desirable item—most often, "revenue." Desirability of an item was based on the item's directional relationship with firm value in common equity valuation models.

TABLE 6

Analysis of the Top 5 Percent of Documents (n = 3,780) Ranked on $Tone_{GI}$–$Tone_{FD}$

Panel A: The Ten Stem Words Included in the GI Positive Wordlist, But Not the FD Positive Wordlist, Which have the Highest Document Frequency

Stem Word	Document Frequency	Word Frequency	*idf* Weighting for n = 75,599
share	3,731	51,774	0.013
compan	3,595	32,980	0.052
actual	3,366	5,488	0.097
common	3,142	28,572	0.209
call	2,953	12,853	0.187
consolid	2,819	10,562	0.236
interest	2,819	28,581	0.240
primaril	2,458	8,551	0.391
discuss	2,444	3,808	0.419
outstand	2,374	7,300	0.418

Panel B: The Ten Stem Words Included in the FD Negative Wordlist, But Not the GI Negative Wordlist, Which have the Highest Document Frequency

Stem Word	Document Frequency	Word Frequency	*idf* Weighting for n = 75,599
risk	3,546	15,548	0.106
uncertain	3,025	6,586	0.268
decreas	2,765	20,543	0.422
lower	1,817	6,781	0.681
declin	1,808	6,894	0.777
less	1,642	4,739	0.810
neg	1,143	2,249	1.225
difficult	848	1,152	1.310
challeng	816	1,126	1.434
least	792	1,032	1.733

We identify the top 5 percent of documents (n = 3,780) ranked on $Tone_{GI}$–$Tone_{FD}$. We then collapse word frequencies in each document into stem word frequencies for the purpose of this analysis. For example, the FD wordlist contains the words UNCERTAINTIES and UNCERTAINTY. If a document contains five occurrences of the word UNCERTAINTY and three occurrences of the word UNCERTAINTIES, then the stem word count for UNCERTAINT is eight. Stem words are created using the PERL module Linquia::Stem for each word in all wordlists. To construct Panel A, we identify all stem words that are included in the GI positive wordlist, but not in the FD positive wordlist. We then identify the ten stem words with the highest overall document frequency in the subsample. For each stem word, the panel includes Document Frequency (number of documents in the subsample containing the word), Word Frequency (total occurrences in the subsample of the stem word), and *idf* Weighting (as developed for the total sample n = 75,599). Panel B is constructed in a similar way, but here we first identify all stem words that are contained in the FD negative wordlist, but not in the GI negative wordlist.

Overall, this analysis supports our conjecture and statistical analysis that researchers are on much firmer ground if they use a domain-specific wordlist. General wordlists are much more likely to include words with meanings that do not match the context of financial disclosure, or to exclude words that would be viewed as positive or negative in a financial disclosure.

Analysis of Post-Earnings Announcement Drift

We analyze post-earnings announcement drift to assess whether tone reflects biased disclosure that initially "fools" investors into mispricing the information, or tone conveys qualitative information about the firm's actual underlying performance. While we cannot distinguish between these two alternatives with certainty, a drift analysis can provide relevant

TABLE 7

Analysis of the Top 5 Percent of Documents
(n = 3,780)
Ranked on $Tone_{FD}$–$Tone_{GI}$

Panel A: The Ten Stem Words Included in the FD Positive Wordlist, But Not the GI Positive Wordlist, Which have the Highest Document Frequency

Stem Word	Document Frequency	Word Frequency	*idf* Weighting, for n = 75,599
increas	3,670	56,630	0.041
growth	3,145	17,968	0.303
strong	2,696	7,672	0.560
certain	2,688	9,838	0.355
higher	2,148	10,392	0.617
record	1,965	6,503	0.922
lead	1,758	2,980	0.759
deliv	1,594	2,937	1.119
expand	1,502	2,618	1.150
grow	1,442	2,332	1.147

Panel B: The Ten Stem Words Included in the GI Negative Wordlist, But Not the FD Negative Wordlist, Which have the Highest Document Frequency

Stem Word	Document Frequency	Word Frequency	*idf* Weighting, for n = 75,599
cost	3,578	37,644	0.098
tax	3,577	43,336	0.113
expens	3,201	22,726	0.227
differ	3,150	4,857	0.145
press	3,110	8,107	0.214
exclud	2,881	22,397	0.395
servic	2,845	23,870	0.384
capit	2,697	12,391	0.300
loss	2,646	26,018	0.233
charg	2,382	18,535	0.551

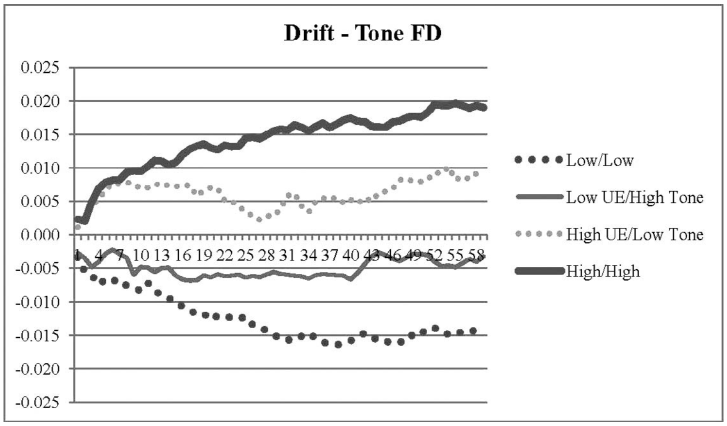
We identify the top 5 percent of documents (n = 3,780) ranked on $Tone_{FD}$–$Tone_{GI}$. We then collapse word frequencies in each document into stem word frequencies for the purpose of this analysis. For example, the FD wordlist contains the words UNCERTAINTIES and UNCERTAINTY. If a document contains five occurrences of the word UNCERTAINTY and three occurrences of the word UNCERTAINTIES, then the stem word count for UNCERTAINT is eight. Stem words are created using the PERL module Linquia::Stem for each word in all wordlists. To construct Panel A, we identify all stem words that are contained in the FD positive wordlist, but not in the GI positive wordlist. We then identify the 20 stem words with the highest overall document frequency in the subsample. For each stem word, the panel includes Document Frequency (number of documents in the subsample containing the word), Word Frequency (total occurrences in the subsample of the stem word), and *idf* Weighting (as developed for the total sample n = 75,599). Panel B is constructed in a similar way, but here we first identify all stem words that are contained in the GI negative wordlist, but not in the FD negative wordlist.

evidence. For example, if overly positive qualitative disclosure initially causes mispricing, then we would expect an initial overreaction to the earnings disclosure followed by drift in the opposite direction of the tone, or, at least, significantly diminished drift (offsetting the typical drift observed after earnings announcements). Alternatively, if tone appropriately reflects the information content of qualitative disclosure, including managers' actual assessment of current and future performance, then we would expect the drift to be highest for firms with both high earnings surprises and positive tone.

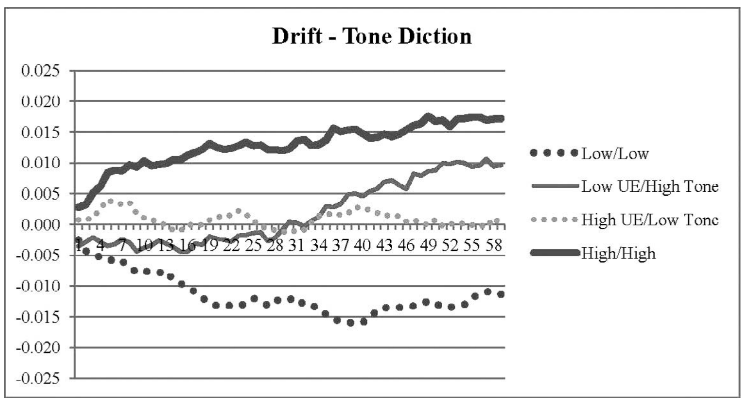
Figure 2 presents the post-earnings announcement drift for four different portfolios, formed as a combination of Good or Bad News and Positive or Negative Tone, where tone is measured using each of the alternative tone measures. We observe the most negative drift for firms with Bad News and Negative Tone (Low/Low) and the most positive drift for firms with Good News and Positive tone (High/High) for three of the alternative tone measures (Figure 2, Panels A, B, and D for $Tone_{FD}$,

FIGURE 2
Post-Earnings Announcement Drift

Panel A: Post-Earnings Announcement Drift for Four Portfolios, with $\Delta Tone_{FD}$



Panel B: Post-Earnings Announcement Drift for Four Portfolios, with $\Delta Tone_{DICTION}$



Panel C: Post-Earnings Announcement Drift for Four Portfolios, with $\Delta Tone_{GI}$

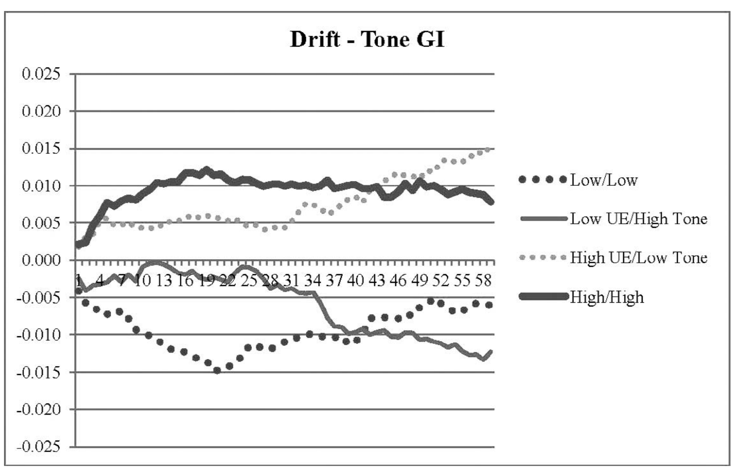


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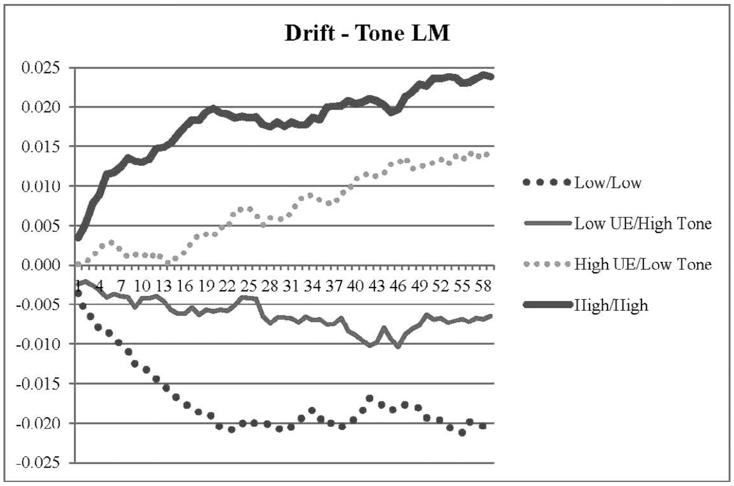
FIGURE 2 (continued)

Panel D: Post-Earnings Announcement Drift for Four Portfolios, with $\Delta Tone_{LM}$



Figures show the post-earnings announcement drift for four portfolios: (1) Low/Low: Bad News at earnings announcement and Negative Tone (bottom quintile in *UE,* bottom quintile in $\Delta Tone_j$); (2) Low UE/High Tone: Bad News at earnings announcement and Positive Tone (bottom quintile in *UE,* top quintile in $\Delta Tone_j$); (3) High UE/Low Tone: Good News at earnings announcement and Negative Tone (top quintile in *UE*, bottom quintile in $\Delta Tone_j$); (4) High/High: Good News at earnings announcement and Positive Tone (top quintile in *UE,* top quintile in $\Delta Tone_j$).
The full-color version of Figure 2 is available as a downloadable PDF file, see Appendix B.

$Tone_{DICTION}$, and $Tone_{LM}$, respectively). However, the direction of the portfolios' drift is consistent with the direction of earnings surprise coupled with direction of tone only for the domain-specific tone measures, $Tone_{FD}$ and $Tone_{LM}$. When the tone measure is either $\Delta Tone_{FD}$ or $\Delta Tone_{LM}$, both Good News portfolios show positive drift with even higher returns when coupled with Positive Tone, and both Bad News portfolios show negative drift with even lower returns when coupled with Negative Tone. These findings suggest that, at least for the domain-specific measures, tone appears to capture qualitative information and managers' objective evaluation of current and future performance.

V. WORD-FREQUENCY TONE MEASURES VERSUS MACHINE-LEARNING TONE MEASURES: EMPIRICAL ANALYSIS

We compare the efficacy of word-count tone measures with machine-learning measures.[25] These measures are created in the following steps: manual classification of a training set of forward-looking sentences from MD&A disclosure into one of four tone categories (positive, neutral, negative, uncertain); application of the Naïve Bayesian classifier algorithm to classify a sample of forward-looking sentences from a larger sample of MD&A disclosures into three categories (positive, neutral, or negative/uncertain); and creation of a tone measure as the average of the tone categories of all forward-looking sentences in a filing.

We start with the Li (2010) 10-K/10-Q sample of 281,433, which spans 1995–2007.[26] For this sample, we extract the MD&A disclosures and retain all sentences that we classify as Forward-Looking Statements (FLS), following Li's (2010) methodology.[27] For each filing's FLS, we compute a word-frequency tone measure using the FD wordlist and equal weighting.

[25] The machine-learning tone measures used in Li (2010) were provided by the author.
[26] Our initial sample differs from Li (2010) because we exclude 1994 filings. At this time, EDGAR filing was not fully implemented and we excluded this year to avoid potential self-selection bias.
[27] To validate our identification of FLS, we compared the number of FLS we identified in each filing to the number of FLS identified by Li (2010). The counts are virtually identical and the correlation between Li's (2010) count and ours is 0.98. See Li (2010, Appendix B) for details on how forward-looking statements are identified.

Data are then merged with Compustat and CRSP. Excluding filings not in CRSP or Compustat and filings without sufficient data for base regressions reduces the sample to 104,863.

Our first analysis of the MD&A disclosures considers the determinants of tone. Similar to Li (2010), the following regression is estimated:

$$TONE^{*} = \alpha + \beta_1 EARN + \beta_2 RET + \beta_3 ACC + \beta_4 SIZE + \beta_5 MTB + \beta_6 RETVOL + \beta_7 EARNVOL + \varepsilon \quad (3)$$

where $TONE^{*}$ is alternately Li's (2010) machine-learning tone measure $Tone_{machine}$ and $Tone_{FD}$, the word-frequency tone measure using the FD wordlist and equal weighting. $Tone_{machine}$ is the average tone of the forward-looking sentences in each firm-quarter, where each sentence's tone has a value of 1, 0, or −1 if classified by the Bayesian learning algorithm as positive, neutral, or negative/uncertain, respectively. Independent variables are computed exactly as in Li (2010):

EARN = quarterly earnings scaled by total assets;
RET = cumulative contemporaneous CRSP stock returns in the fiscal quarter;
ACC = quarterly earnings minus quarterly cash from operations scaled by total assets;
SIZE = logarithm of the market value of equity at the end of the quarter;
MTB = market value of equity plus the book value of total liabilities scaled by total assets;
RETVOL = standard deviation of monthly stock returns in the 12 months prior to the fiscal quarter ending date; and
EARNVOL = standard deviation of quarterly earnings in the previous five years.

EARN, RET, ACC, SIZE, MTB, RETVOL, and *EARNVOL* are winsorized the top and bottom 1 percent. Regressions also include firm and quarter fixed effects.

Our regression results, shown in Table 8, indicate that both $Tone_{machine}$ and $Tone_{FD}$ are fairly well explained by fundamentals. In addition, both $Tone_{machine}$ and $Tone_{FD}$ are positively associated with *EARN,* after controlling for *RET, ACC, SIZE, MTB, RETVOL,* and *EARNVOL*. These findings are consistent with Li (2010).[28] Although the coefficient on *EARN* is significant and positive in both models, the control variables appear to do a better job of explaining $Tone_{machine}$ compared to $Tone_{FD}$. The R^2 in the $Tone_{FD}$ regression (column 1) is only 49 percent, compared to 65 percent in the $Tone_{machine}$ regression (column 2). A Vuong (1989) test of differences is significant at $p < 0.01$.

For our analysis of the relation between the alternative tone measures and firms' future profitability, we regress future earnings on the alternative tone measures and the relevant control variables defined in Li (2010):

$$Earn_{t+1} = \alpha + \beta_1 TONE + \beta_2 EARN + \beta_3 RET + \beta_4 ACC + \beta_5 SIZE + \beta_6 MTB + \beta_7 RETVOL + \beta_8 EARNVOL \quad (4)$$

where all variables are as defined above, and $Earn_{t+1}$ equals the earnings in the next quarter scaled by the book value of assets at the end of the current quarter. The regression also includes firm/quarter fixed effects. As shown in Table 8, the coefficients on $Tone_{FD}$ and $Tone_{machine}$ are both significant, with t-statistics of 3.14 and 4.69, respectively. These results indicate that regardless of whether tone is measured using word-frequency counts or a machine-learning classification algorithm, when the tone of the MD&A is more positive, the earnings in the following quarter are significantly higher.[29] As further support that the two measures have similar explanatory power, the R^2s of the two regressions are virtually identical (71.0 percent versus 71.2 percent) and a Vuong (1989) test of differences cannot reject the null that the R^2s are equivalent.

In summary, results indicate that the relation between MD&A tone and future earnings is statistically significant regardless of whether the tone measure is derived from word-frequency counts or from the machine-learning classifier algorithm that has been used in capital markets research. At least in the context of MD&A disclosure, there appears to be very little difference in the power of tests using $Tone_{FD}$ versus $Tone_{machine}$ as alternative proxies for tone.

VI. CONCLUSION

In this study, we evaluate methods used in accounting and finance to quantify the tone of financial disclosure. Overall, our findings suggest that capital markets researchers aiming to measure the tone of financial disclosure (1) can increase the power of their short-window tests and avoid Type II errors, particularly when sample size is restricted, by using a domain-specific wordlist (e.g., FD or LM) and a change rather than level specification of tone; (2) can create replicable results almost

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[28] All results for $Tone_{machine}$ are similar to those in Li (2010) with the exception of *ACC* (which was significantly negative) and *MTB* (which had the opposite sign). These differences are likely explained by the fact that we winzorize all continuous variables at ±1 percent, exclude filings containing more than 100 FLS, and exclude 1994 filings.

[29] In related research, Davis and Tama-Sweet (2010) find that tone of MD&A disclosure measured as in Henry (2006), i.e., $Tone_{FD}$, is positively associated with future firm performance (measured as return on assets), while tone measured using Diction has a negative relation with future firm performance.

TABLE 8

Tone of Forward-Looking Statements in MD&A Disclosure and Future Performance Comparison of Machine Learning Measure to Domain-Specific Word-Frequency Measure (n = 104,863)

	Dependent Variable: *Tone*		Dependent Variable: $Earn_{t+1}$	
	FD Coefficient (t-stat)	Machine Learning Coefficient (t-stat)	FD Coefficient (t-stat)	Machine Learning Coefficient (t-stat)
*Tone**			0.002***	0.005***
			(3.14)	(4.69)
EARN	0.128***	0.037***	0.357***	0.357***
	(5.43)	(2.78)	(99.03)	(99.06)
RET	−0.002	0.015***	0.007***	0.007***
	(−0.40)	(6.31)	(11.56)	(11.45)
ACC	0.057***	−0.002	−0.101***	−0.101***
	(3.68)	(−0.017)	(42.54)	(42.51)
SIZE	0.022***	−0.004***	0.006***	0.006***
	(11.48)	(−3.34)	(19.33)	(19.53)
MTB	0.013***	0.002***	0.001***	0.001***
	(12.15)	(2.52)	(7.68)	(7.78)
RETVOL	−0.193***	−0.164***	−0.022***	−0.022***
	(−11.18)	(−16.77)	(−8.41)	(−8.41)
EARNVOL	0.136***	−0.034**	0.019***	0.019***
	(5.87)	(−2.51)	(5.42)	(5.42)
Firm/Quarter Fixed Effects	Yes	Yes	Yes	Yes
R^2	49.0%	64.9%###	71.0%	71.2%
n	104,863	104,863	104,863	104,863

***, **, * Indicate significance at < 0.001, < 0.01, and < 0.05, respectively, two-tailed.
Vuong tests of differences are significant at ($p < 0.01$) when *Tone* is the dependent variable (first two columns), but insignificant in the earnings prediction models (last two columns).
*Tone** is alternately Li's (2010) machine-learning tone measure $Tone_{machine}$ and $Tone_{FD}$, the word-frequency tone measure using the FD wordlist and equal weighting. *EARN, RET, ACC, SIZE, MTB, RETVOL,* and *EARNVOL* are winsorized at the top and bottom 1 percent.

Variable Definitions (numbers below preceded by # are the Compustat quarterly data items):
$Earn_{t+1}$ = earnings in the next quarter (#69) scaled by the book value of assets at the end of the current quarter (#44). Other independent variables are computed exactly as in Li (2010);
EARN = quarterly earnings scaled by total assets (#69/#44);
RET = cumulative contemporaneous CRSP stock returns in the fiscal quarter;
ACC = quarterly earnings minus quarterly cash from operations scaled by total assets [(#69 − #108)/#44]. Since item 108 is cumulative cash flows, current quarter cash flows are computed by differencing current and previous quarters' values in quarters 2–4;
SIZE = the logarithm of the market value of equity at the end of the quarter (#14 ∗ #61);
MTB = market value of equity plus the book value of total liabilities scaled by total assets [(#14 ∗ #61 + #54)/#44];
RETVOL = standard deviation of monthly stock returns in the 12 months prior to the fiscal quarter ending date; and
EARNVOL = standard deviation of quarterly earnings in the previous five years.

exclusively by using equal weighting rather than *idf* weighting; and (3) can use word-count tone measures to achieve results that are virtually equivalent to at least one machine-learning tone measure in tests of the relationship between MD&A tone and future earnings. Equal-weighted word-frequency measures produce both powerful and replicable results in commonly studied settings for measuring disclosures' qualitative information in capital markets research.

This paper focuses on the tone of two specific types of corporate disclosure: earnings announcements and the MD&A. One potentially fruitful avenue for future capital markets research on this topic is to address trends and variations in the wordlists best suited to aggregating qualitative information in disclosures and in other types of financial narrative (e.g., analysts' reports and financial journalists' discussion of company performance). Additional areas for future research are the study of tone in post-earnings announcement drift and the impact of tone on stakeholders other than investors.

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APPENDIX A
Document-Weightings

This appendix presents examples of document-weighting, specifically, the *idf* (inverse document frequency), in developing measures of qualitative information. Two of the basic tasks in computational linguistics are content analysis and information retrieval, i.e., search. To find or retrieve a document of interest from among a corpus, some measure of the content of each document in the corpus is combined with a weighting that identifies the document most likely to be relevant for the search. Measures of content are typically based on proportional word frequency. Weightings used to improve the precision of a search by ranking documents within a corpus according to relevance are typically based on frequency of word occurrence across all documents in a corpus; less frequently occurring words can better rank the relevance of documents that are related to a specific search term.

Idf weightings are designed to favor words that occur in fewer documents because those words improve the ability to rank documents according to their relevance to a particular set of search words and, thus, to improve the precision of a search algorithm. The *idf* weight is defined as $\log(N/n_j)$, where N is the total number of documents and n_j is the number of documents that contain the search word. A search word that appears in many documents would contribute little to ranking documents' relevancy to the search, and a search word that appeared in every document in the corpus would contribute nothing; its *idf* weight would be $\log(1) = 0$. Thus, *idf* weightings are dependent on sample selection.

The *wf-idf* weight combines a word's frequency count within each document and a measure of its frequency across all documents in a corpus (its inverse document frequency) into a single weight. For each word i in a document j, the *wf-idf* weight is the product of its word frequency *wf* weight times the *idf* weight. The *wf-idf* measure can be formed as shown below (from Manning and Schütze 2002, 543).

$$wf\text{-}idf = \begin{cases} \left(1 + log(wf_{i,j})\right) log(N/n_j) & \text{if } wf_{i,j} >= 1 \\ 0 & \text{if } wf_{i,j} = 0 \end{cases} \tag{A1}$$

The first example in this appendix illustrates how *idf* weights make tone measures extremely sensitive to the sample corpus. Assume a hypothetical set of three documents, each with a total of 100 words, with the word-frequency count matrix shown in Panel A of Table 9. (For simplicity, our examples use documents of identical lengths.) In a word-frequency count matrix, documents are represented as vectors of word frequency counts. For example, the word "less" appears ten times in Document 1 (D1), and the word "lower" appears ten times in Document 2 (D2) and one time in Document 3 (D3). The words "less" and "lower" are the only two negative words in the corpus. The remaining words in the document would each be represented as a separate column of the matrix, but for presentation purposes, we show them here in a single column labeled "another." The *idf* weight for each word is presented at the bottom of the column. The *idf* weight for the word "less" is 1.10 because it appears in only one of the three documents, ($idf = \ln(3/1) = 1.10$. The *idf* weight for the word "lower" is only 0.41, calculated as ($idf = \ln(3/2)$), because it appears in two of the three documents.

Two NEG measures for each of the three documents are shown in Panel B of Table 9, the first measure using equal weighting and the second measure using the combined *wf-idf* weighting according to Equation (A1), above. Because D1 and

TABLE 9

Sensitivity of *idf*-Weighted Tone Measures to Composition of Sample Corpus

Panel A: Word-Frequency Count Matrix for Set of Three Hypothetical Documents

	"LESS"	*"LOWER"*	*"Another"*	**Total Words**
D1	10	0	90	100
D2	0	10	90	100
D3	0	1	99	100
idf weight (N/n_j)	1.10	0.41	0.00	

Panel B: NEG, Using Equal Weighting versus *idf*-Weighted Measures

	NEG, Measured As:	
	Equal-Weighted *wf* Negative Words as a Proportion of Total Words	**Inverse-Document-Frequency-Weighted *wf-idf* (Equation A1)**
D1	0.100	3.628
D2	0.100	1.339
D3	0.010	0.405

D2 have the identical proportion of negative words, the equal-weighted measure of NEG is identical for the two documents. When the NEG measure adds an *idf* weighting, D1 is scored as 2.7 times more negative than D2, but the increase in D1's NEG measure is solely because the word "lower" appears once in D3. If the corpus had excluded D3 or if D3 had not contained the word "lower," then the *wf-idf* NEG measure for D2 would have been identical to that of D1.

The next example illustrates how the *idf* weights serve to correct for errors in mistakenly classified words that commonly occur across a corpus. Table 10, Panel A shows the word-frequency count matrix for a hypothetical three-document corpus. Two alternative wordlists are used to create NEG measures for the documents. The first wordlist includes only two negative words, "less" and "lower," while the second wordlist incorrectly classifies the word "division" as negative. Based on the equal-weighted measures shown in Panel B, the incorrect wordlist scores the documents as far more negative than the correct wordlist (four times more negative than the correct wordlist for D1 and D2, and 31 times more negative for D3). An *idf* weighting serves to negate the effect of the incorrectly classified word because it is very common in the corpus. Specifically, because the incorrectly classified word "division" appears in all three documents, its *idf* weight is 0 ($idf = \log(3/3)$), so the *wf-idf* measure eliminates the word from the negative tone measure. Using the *wf-idf* measure, therefore, the incorrect wordlist can produce exactly the same NEG measure as the correct wordlist. Of course, correcting the list by removing the improperly classified word "division" from the negative wordlist would be a simpler and more direct solution.

In summary, although *idf* weights can correct tone measures for misclassified words that appear frequently in a corpus, a far more direct solution would be simply to remove the mistakenly classified words from the wordlist. In addition, *idf* weights do not fully solve the problem of misclassification. First, adding *idf* weights does not serve to include negative words that were mistakenly omitted from a wordlist. Second, when a misclassified word occurs infrequently across a corpus, *idf* weights exacerbate their impact on the NEG measure. Thus, use of *idf* weights potentially offers only limited benefits, but comes with a significant cost: in addition to being more difficult to implement, it reduces the transparency and replicability of research using word-count measures of qualitative information.

TABLE 10

***Idf* Weights' Down-Weighting of Erroneously Classified Words that Appear Frequently in a Corpus**

Panel A: Word-Frequency Count Matrix for Set of Three Hypothetical Documents

	"LESS"	*"LOWER"*	*"DIVISION"*	***Another***	**Total**
Included in correct negative list	Yes	Yes	No	No	
Included in incorrect negative list	Yes	Yes	Yes	No	
D1	10	0	30	60	100
D2	0	10	30	60	100
D3	0	1	30	69	100
idf weight (N/n_j)	1.10	0.41	0.00	0.00	

Panel B: NEG Measures, Using Equal-Weighted and *idf*-Weighted Measures Applied to the Correct Wordlist versus the Incorrect Wordlist

	NEG, Measured as:					
	Equal-Weighted *wf* Negative Words as a Proportion of Total Words			**Inverse-Document-Frequency-Weighted *wf-idf* (Equation A1)**		
	Measure Using Correct List	**Measure Using Incorrect List**	**Ratio of Incorrect to Correct**	**Measure Using Correct List**	**Measure Using Incorrect List**	**Ratio of Incorrect to Correct**
D1	0.100	0.400	4	3.628	3.628	1
D2	0.100	0.400	4	1.339	1.339	1
D3	0.010	0.310	31	0.405	0.405	1

APPENDIX B

Figures 1 and 2: http://dx.doi.org/10.2308/accr-51161.s01